Eldin Lozic

San Jose, CA | U.S. Citizen
408-529-5154 | eldinlozic1@gmail.com
linkedin.com/in/eldin-lozic-20781838a | github.com/EldinLz

Education

San Jose State University (SJSU) Expected Dec. 2027

B.S. Electrical Engineering | Overall GPA: 3.83/4.00

Coursework: Electronic Design I, Microprocessor-Based System Design, Physical Electronics, Probability and Statistics in EE (in progress); Circuits and Systems, Signal Processing, Digital Logic Circuit Design (completed).

Technical Skills

Electronics: KiCad schematic/PCB design, LTspice, circuit analysis, schematic/datasheet review, power/current budgeting, oscilloscope, multimeter, soldering, breadboarding, fault isolation, ERC/DRC.

Embedded and Analysis: C, STM32L476, GPIO/register-level programming, I2C, Arduino, Python (NumPy, SciPy, pandas, Streamlit, Plotly), MATLAB.

Digital Design: Verilog, Vivado, combinational/sequential logic, simulation, testbenches.

Engineering Experience

EL Electric | Apprentice / Assistant Electrician Jan. 2021 - Present

- Installed and terminated 120 V/240 V residential circuits, including switches, receptacles, lighting, and panel connections.
- Diagnosed shorts, open circuits, overloads, and grounding faults with voltage, resistance, and continuity measurements; assisted with subpanel and GFCI/AFCI checks under licensed supervision.

Engineering Projects

ESP32 Sensor Data Acquisition Board | KiCad, ESP32-C3, Arduino

- Designed a two-layer KiCad ESP32-C3 board with an MCP6002-buffered photoresistor ADC, BME280 I2C sensor, status LED, and six labeled test points.
- Completed schematic ERC and PCB DRC, generated Gerber/drill files, and hand-soldered and assembled Rev. A hardware.
- Loaded Arduino firmware and verified 3.3 V power, serial CSV output, environmental sensing, and light response from approximately 6% to 100% of ADC full scale.

STM32L476 Embedded Systems | C, GPIO, I2C

- Configured GPIO and peripherals in C for LED/button control, stepper-motor sequencing, keypad input, and OLED display interfaces.
- Debugged timing and I/O behavior through incremental bench testing and register-level troubleshooting.

VibeGuard Motor Condition Monitor | Python, Streamlit, PCB Design

- Built a Python signal-analysis application with waveform/FFT analysis, ADC/filter simulation, synchronized vibration/current visualization, CSV import, and report export; verified behavior with 19 automated tests.
- Designing an ESP32-S3 carrier PCB for ADXL345 vibration sensing, INA219 current monitoring, and PWM motor control.

Spartan Racing BSPD Electronics | Formula SAE

- Analyzed shutdown-system schematics and component datasheets to estimate nominal current draw and a conservative 5 V power budget.
- Calculated worst-case relay-coil current from tolerance data and documented assumptions and open verification items for team power allocation.